

Talh Hossin

AI Specialist

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SUMMARY

Senior Computer Science student with a focus on Artificial Intelligence, Machine Learning, and Software Engineering. Skilled in developing AI-powered web applications and training deep learning models through hands-on projects. Experienced in creating intelligent systems that address challenges in healthcare, accessibility, and automation. I work at the intersection of AI and cybersecurity, designing systems that are not only intelligent but also secure and adaptive. What drives me is understanding how machines learn, reason, and make decisions in ways that resemble human cognition. I believe AI will transform every industry, and I am committed to contributing to that future.

EDUCATION

Bachelor of Science in Computer Science (AI Concentration)

Effat University

09/2023 – 06/2027

Jeddah, KSA

SKILLS

- **Programming:** Python, Java, C++, JavaScript
- **Web Development:** HTML, CSS, Bootstrap, PHP
- **AI & ML:** PyTorch, TensorFlow, HuggingFace, NLP, Adversarial ML
- **Data Tools:** Pandas, NumPy, Matplotlib, Scikit-learn
- **Systems & Tools:** Linux, Git, Jupyter, Cisco Packet Tracer, MySQL
- **Soft Skills:** Problem Solving, Research, Communication, Teamwork, Time Management

LANGUAGES

Arabic

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English

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French

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German

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CERTIFICATES

2025 🔗

Python for Data Science, AI & Development : IBM

2025 🔗

Introduction to Artificial Intelligence (AI): IBM

2023 🔗

Python Data Structures:

2023 🔗

Digital Safety and Security

AWARDS

The 23rd International Learning and Technology Conference (L&T 2026)

2026

Effat University / IEEE

AI-Driven Cognitive Health Monitoring: Early Dementia Detection via Social Media Bio markers. Showcased an NLP-based system using ClinicalBERT and big-data architecture for early dementia prediction through linguistic analysis of Arabic social-media text.

Global Occupational Safety and Health Hackathon (GOSHATHON)

2025

Effat University / GOSHATHON Organizing Committee

Proposed an AI-driven predictive risk intelligence platform for high-risk workplaces. Designed machine learning predictive risk models, temporal analysis, and context-aware scoring to detect hazards before incidents occur. The solution aligns with Saudi Vision 2030, promoting preventive safety and digital transformation in industrial environments.

Active Member in IEEE CS Club

Effat University Events

Volunteered throughout my BSc in events yearly and served as an active member of the IEEE Computer Society Club, contributing to university conferences, technical activities, and student support initiatives.

PROJECTS

Cognitive Health Monitoring System <i>Detecting Cognitive Decline (Dementia Detection)</i> Built an AI-driven tool using ClinicalBERT and NER to detect early dementia indicators from social-media text. Integrated UML design and scalable architecture for multilingual, culturally adaptive healthcare models. Applied Arabic text analysis for inclusive AI development.	2026
Data Driven Insights for Emergency Triage <i>Exploratory Data Analysis & Visualization for Triage Prediction</i> Designed a Data Science workflow for predicting emergency-room triage levels through advanced data cleaning and exploratory data analysis (EDA). Applied visualization techniques to uncover medical-urgency patterns, achieving 87% accuracy and enabling reliable triage predictions in high-stakes healthcare environments.	2026
Dementia Sentiment Analysis & Data Visualization <i>Arabic Data Cleaning, Sentiment Analysis, and Ethical Visualization</i> Collected a 5-year Arabic dataset from real dementia patients on Facebook. Professionally cleaned and reconstructed scattered, noisy text into a reliable, analysis-ready dataset. Applied Arabic text preprocessing and sentiment analysis to extract emotional and linguistic patterns related to cognitive decline. Created meaningful visualizations to communicate insights while ensuring strict data security and ethical handling.	2026
Smart Triage  <i>Patient Condition Monitoring AI</i> Developed an intelligent system that predicts clinical urgency from real-time vitals using advanced Machine Learning and Healthcare AI. Engineered for high-stakes environments with concept-drift handling and data-integrity validation to ensure reliability under evolving hospital data. Achieved 94.5% accuracy and demonstrated robust performance in prioritizing life-threatening cases through adaptive model testing.	2026
Tourist Trip Planner <i>Optimized Route Planning and Multi-Mode Travel Analysis</i> Developed a C++ and Python application for optimized travel route planning using graph algorithms and Dijkstra's pathfinding. Implemented multi-mode transport selection, cost estimation, and collaborative trip planning features. Designed efficient CSV-based data handling for route storage and retrieval, ensuring scalability and performance in real-world travel scenarios.	2024